

PRANAY MESHARAM

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EDUCATION

• Indian Institute of Science Education and Research Bhopal

B.S. in Data Science and Engineering

2023 – Present

Bhopal, India

PUBLICATIONS

• **COMBAT: Conditional World Models for Behavioral Agent Training**

Anmol Agarwal, Pranay Meshram, Sumer Singh, Saurav Suman, Andrew Lapp, Shahbuland Matiana, Louis Castricato, Spencer Frazier

International Conference on Computer Vision (ICCV) Workshop on Reliable and Interactive World Models (October 2025)

EXPERIENCE

• Research Intern

Indian Institute of Technology Roorkee

July 2025 – Present

Remote

- Developed novel methods for Indoor Visual Place Recognition.
- Led large-scale 2D/3D dataset generation using **Infinigen** and **Flux-Fill**.

• Summer Research Intern

Multi-Agent Simulation & Control Lab, IISER Bhopal

May 2025 – July 2025

On-site

- Implemented and benchmarked SOTA open-vocabulary segmentation models (**ZegCLIP**, **ZegFormer**, **CAT-Seg**).
- Conducted a literature review on open-vocabulary segmentation and multi-modal learning.

PROJECTS

• Tekken-DCAE

Deep Compression Autoencoder mapping Tekken 3 clips to probabilistic latents for a 1.2B-parameter world model



- Built a **340M-parameter** model with a 4-channel (RGB + pose-map) pipeline to jointly embed agent kinematics and visual semantics at **32× compression**.
- Tuned a hybrid loss (**weighted LPIPS + MSE + KL**) to minimize perceptual blur.
- Scaled multi-GPU training on **8× H200** with optimized throughput and memory.

• CLINO-VPR: CLIP-Guided Semantic Gating for Visual Place Recognition

Dual-encoder VPR model using **DINOv2-L + CLIP ViT-L/14-336** for semantics-driven retrieval



- Designed a lightweight **Semantic Gate** (MLP + sigmoid) that turns the CLIP vector into a spatial mask, suppressing clutter (sky, vegetation) in DINOv2 features.
- Aggregated gated features via **SALAD** (64 clusters) into a 512-d L2-normalized descriptor.
- Achieved **R@1 = 37.37%** at 20 m on the NYC benchmark (**+2.17 pp** over SALAD baseline); trained on GSV-Cities with MultiSimilarity loss.

• SigLIP-Seg

Cost-aggregation framework for open-vocabulary segmentation with a SigLIP encoder



- Adapted CAT-Seg to a SigLIP backbone and trained on COCO-Stuff.
- Built a custom preprocessing and evaluation pipeline for large-scale segmentation experiments.

• Occupancy Estimation from Thermal Imagery

Real-time occupancy detection on edge hardware



- Built a custom thermal dataset (**1,000+ samples**) using a Raspberry Pi + MLX90640 sensor.
- Trained a **YOLOv8** detector with augmentation and quantization for edge deployment.
- Deployed real-time inference on Raspberry Pi for lightweight, privacy-preserving monitoring.

• 4-bit LLaMA2-7B

Fine-tuned LLaMA2 on the Puffin dataset for creative text generation



- Applied **QLoRA** (LoRA + 4-bit quantization via BitsAndBytes) to fine-tune on a single Colab GPU.
- Reformatted a custom Hugging Face dataset to match the LLaMA2 prompt template.

SKILLS

- **Programming Languages:** Python, C, C++, MATLAB, Wolfram, JavaScript, SQL, Bash
- **ML Frameworks:** PyTorch, Hugging Face Transformers, PEFT, Diffusers, BitsAndBytes, Scikit-Learn, OpenCV
- **Tools & Infrastructure:** Git, Docker, Linux, Slurm, WandB, Jupyter, VS Code, Vim
- **Research Interests:** Vision-Language Models, World Models, Generative Modeling (VAE/Diffusion), Visual Place Recognition, Open-Vocabulary Segmentation